

Md. Mahmudur Rahman

AI Engineer (LLMs & Agentic Systems)

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Profile

Production-grade **AI Engineer** specializing in autonomous agentic frameworks including **OpenClaw**, **Hermes**, and **LangGraph**, multi-agent orchestration, and advanced Retrieval-Augmented Generation (RAG) pipelines. Expert in building asynchronous **Python/FastAPI** microservices, developing **n8n automation tools**, and rapid full-stack agentic/vibe coding for real-time state streaming. Proven track record deploying AI architectures on **AWS EC2 and S3**, vLLM, and Docker. **NASA Space Apps Global Nominee**.

Technical Skills

ML & Deep Learning: PyTorch, HuggingFace Transformers, SentenceTransformers, scikit-learn, Model Evaluation, FP16 Optimization
RAG & Vector Databases: FAISS, ChromaDB, BM25 Retrieval, Reciprocal Rank Fusion (RRF), Hybrid RAG, Knowledge Tracing
Backend & APIs: Python, FastAPI, Pydantic, WebSockets, REST APIs, Redis, MongoDB, PostgreSQL, Playwright
Automation & Frontend: n8n Workflow Automation, Custom n8n Tool Nodes, Agentic Vibe Coding, React 18/19, TypeScript
Cloud, DevOps & MLOps: AWS (EC2, S3), Docker, Docker Compose, Git, Linux, HuggingFace Spaces, Render, CI/CD
Agentic AI & Frameworks: OpenClaw, Hermes, LangGraph, LangChain, CrewAI, AutoGen, LoRA/QLoRA Fine-Tuning, vLLM, Groq

Experience

AI Developer (Multi-Agent Systems)

Jul 2026 – Present

Studio Butterfly

Dhaka, Bangladesh

- Architect and deploy agentic AI products, multi-agent graph models (research, planning, execution), and LLM features for SaaS platforms using **LangGraph** and **LangChain**.
- Develop production Retrieval-Augmented Generation (**RAG**) pipelines and vector database integrations for high-precision semantic search and retrieval.
- Engineer structured LLM outputs, active context-window management, and output validation schemas to minimize hallucinations and agent failure rates.
- Build Agent-to-Agent (**A2A**) communication protocols and deploy Model Context Protocol (**MCP**) tool-calling for autonomous agent execution.
- Optimize backend API latency, token overhead costs, and backend scaling parameters for high-throughput SaaS workloads.

AI Engineer

Feb 2026 – Jul 2026

Softvence

Dhaka, Bangladesh

- Architected autonomous agentic workflows using **OpenClaw** and **Hermes** alongside LangGraph models to orchestrate self-correcting business logic.
- Developed custom AI backend solutions using **Python** and **FastAPI** microservices, streaming live outputs over real-time **WebSockets**.
- Managed DevOps and deployment cycles on **AWS EC2/S3** and containerized application runs via **Docker Compose** and **n8n** workflow automation.

Featured Projects

APEX Hybrid AI Lab — Local-First Multi-Agent System | Python, LangGraph, OpenClaw, Hermes Agent



- Architected a secure, local-first multi-agent system featuring a master routing orchestrator, self-healing execution loops, conversational memory, and autonomous tool calling.
- Built resilient fallback mechanisms and execution guardrails to eliminate agent loops, minimize hallucination rates, and handle complex multi-turn user intent flows.

BanglaSupport-LLM — Fine-Tuned Qwen3-8B Architecture | PyTorch, HuggingFace, QLoRA, RAG, FastAPI



- Fine-tuned a Qwen3-8B model using 4-bit QLoRA (NF4) for domain-specific customer support, featuring agentic tool calling and RAG policy retrieval.

- Engineered an asynchronous FastAPI microservice with streaming WebSockets backends and a React UI for real-time inference delivery.

Lumina Analyst — Autonomous AI Data Agent | FastAPI, Groq/OpenAI, pandas, Plotly, React 19, TypeScript



- Engineered an autonomous AI data analysis agent with **LLM function calling** that accepts natural language queries over uploaded datasets, routes to specialized tools, and streams step-by-step reasoning in real time.
- Built an analytical engine with smart schema compression for token budgets and secure Python sandboxed execution; generates interactive Plotly charts with zero manual configuration.

Autonomous AI Research Agent | FastAPI, LangChain, FAISS, BM25, WebSockets, React 18, Docker



- Designed a 6-agent orchestration system (Planner → Search → Paper → Summarizer → Critic → Writer) that autonomously executes literature discovery, PDF parsing, factual verification, and citation synthesis.
- Engineered Hybrid RAG combining FAISS vector search, BM25 keyword matching, and Reciprocal Rank Fusion (RRF); streamed live pipeline updates over WebSockets to a React interface.

Research & Working Papers

EduTutor-LaPIT: Tri-Modal Latent Prefix Fusion for Outcome-Aware Tutoring

Under Preparation

- Architected a novel tri-modal outcome-aware tutoring system fusing **Knowledge Tracing (SimpleKT)**, **Differentiable Soft-RAG**, and **Pedagogical Policy** embeddings into causal LLMs via **Latent Prefix Interface (LPI)** tokens.
- Implemented 4-bit QLoRA fine-tuning infrastructure using BitsAndBytes NF4 and Flash Attention 2 on 8B parameter models; executed 10-seed matrix sweeps with paired Wilcoxon signed-rank and Benjamini-Hochberg FDR statistical significance testing.

Empirical Analysis of SGD Optimization in Deep Architectures

Ongoing Research

- Investigating variance reduction, dynamic learning-rate schedules, and optimization trajectory stability in stochastic gradient descent algorithms applied to quantized neural networks.

Education

Daffodil International University (DIU)

B.Sc. in Computer Science and Engineering; GPA: 3.63 / 4.00

Dhaka, Bangladesh

Jan 2022 – Dec 2025

Honors & Awards

NASA Space Apps Challenge 2024 — Regional Winner & Global Nominee

Oct 2024

- Selected among <50 global finalist teams worldwide out of thousands of international submissions.

NASA Space Apps Challenge 2023 — Regional 2nd Runner-Up

Oct 2023

- Consecutive podium finish across independent global challenges.

Competitive Programming — Top 20 DIU CPC; ICPC Preliminary Contestant

Leadership & Activities

Cyber Security Club, Daffodil International University

Sep 2024 – Jan 2026

Lead Executive

Dhaka, Bangladesh

- Served as a core organizer and committee member for DIU CyberCon - National CTF 2025, the first-ever national Capture the Flag competition hosted at DIU, directing event logistics, managing cross-functional volunteer teams, and orchestrating security operations including LLM jailbreaking and AI security challenges.

Creative IT Institute

Jun 2023 – Jan 2026

Campus Ambassador

Dhaka, Bangladesh

- Represented the institute on campus, coordinating Generative AI workshops, prompt engineering seminars, and registration booths to drive technical training enrollment, student engagement, and brand awareness.

BASIS Student's Forum, DIU Chapter

Nov 2024 – Jan 2026

Executive

Dhaka, Bangladesh

- Supported campus-wide technology initiatives, community outreach programs, and student networking seminars focused on Generative AI development and industry-level tech skills.